

Simple Docker Guide

By the end of this guide, you'll know how to:

- Understand what Docker is
- Install Docker
- Use basic Docker commands
- Run your first container

What is Docker?

- Docker is a tool that makes it easier to run apps in a **container**.
- A container is like a mini-computer inside your computer.
- It includes everything an app needs to run.

 Think of it like a **magic box** that runs programs the same way everywhere.

Why Use Docker?

- ✓ No "it works on my machine" problems
- ✓ Easy to install software
- ✓ Great for learning programming & deploying apps

Installing Docker

Windows & macOS

1. Go to <https://www.docker.com/products/docker-desktop>
2. Download Docker Desktop
3. Install & Restart your computer

Linux (Ubuntu example)

```
sudo apt update  
sudo apt install docker.io  
sudo systemctl start docker  
sudo systemctl enable docker
```

First Test: Hello World

Try this in your terminal:

```
docker run hello-world
```

✓ If it prints a welcome message, Docker works!

Useful Docker Commands

```
docker images      # Show downloaded images
docker ps          # Show running containers
docker ps -a       # Show all containers
docker stop <id>   # Stop a container
docker rm <id>     # Remove a container
docker rmi <image> # Remove an image
```

Run Python in Docker

```
docker run -it python:3
```

➡ This opens a Python shell inside a container. Try using `print("Hello")` .

Type `exit()` to leave.

Dockerfile Example

```
# Save this as Dockerfile
FROM python:3
WORKDIR /app
COPY . /app
CMD ["python", "main.py"]
```

Then build and run:

```
docker build -t myapp .
docker run myapp
```

Summary

- ✓ Docker runs apps in containers
- ✓ Easy to install & use
- ✓ Great for learning and sharing apps

Resources

- docker.com
- docs.docker.com/get-started
- YouTube: "Docker in 5 Minutes"

🎓 Have fun exploring!

Thank You!

 Questions? Ask your TA or search online.
You're on your way to becoming a DevOps pro!

